Number of Questions: 30 No. of Pages: 4

Student Full Name:

Student ID number: Answer Sheet

(غير مسموح باستخدام الآلة الحاسبة في الامتحان)

(30x1=30 Marks)

Choose the correct answer

1. The binary number, $(10101010)_2$ is equivalent to:
   (a) $(206)_8$ (b) $(522)_8$ (c) $(255)_8$ (d) $(252)_8$
2. The binary number, $(1011.1110)_2$ is equivalent to:
   (a) $(11.875)_{10}$ (b) $(11.675)_{10}$ (c) $(13.875)_{10}$ (d) $(13.75)_{10}$
3. The binary number, $(1010000)_2$ is equivalent to:
   (a) $(96)_{10}$ (b) $(80)_{10}$ (c) $(78)_{10}$ (d) $(84)_{10}$
4. The binary number, $(101111011110)_2$ is equivalent to:
   (a) $(2F77)_{16}$ (b) $(4EEE)_{16}$ (c) $(77F2)_{16}$ (d) $(2F7E)_{16}$
5. The decimal number, $(12.125)_{10}$ is equivalent to:
   (a) $(1100.0110)_2$ (b) $(1110.0010)_2$ (c) $(1100.0010)_2$ (d) $(1010.1100)_2$
6. The decimal number, $(5238)_{10}$ is equivalent to:
   (a) $(1327)_{16}$ (b) $(12166)_{16}$ (c) $(1476)_{16}$ (d) $(1388)_{16}$
7. The decimal number, $(31)_{10}$ is equivalent to:
   (a) $(37)_8$ (b) $(43)_8$ (c) $(33)_8$ (d) $(30)_8$
8. The octal number, $(115)_8$ is equivalent to:
   (a) $(77)_{10}$ (b) $(98)_{10}$ (c) $(68)_{10}$ (d) $(74)_{10}$
9. The octal number, $(125)_8$ is equivalent to:
   (a) $(76)_{16}$ (b) $(55)_{16}$ (c) $(65)_{16}$ (d) $(67)_{16}$
10. The hexadecimal number, $(1F5)_{16}$ is equivalent to:
    (a) $(496)_{10}$ (b) $(512)_{10}$ (c) $(288)_{10}$ (d) $(501)_{10}$
11. The addition of binary numbers, $(11011011010)_2 + (010100101)_2$ is equal to:
    (a) $(0111001000)_2$ (b) $(1100110110)_2$
    (c) $(11101111111)_2$ (d) $(10011010011)_2$
12. The subtraction of binary numbers, $(101111)_2 - (010101)_2$ is equal to:
    (a) $(100100)_2$ (b) $(010101)_2$ (c) $(011010)_2$ (d) $(011001)_2$
13. The addition of hexadecimal numbers, $(2F3D)_{16} + (FB1A)_{16}$ is equal to:
    (a) $(12A57)_{16}$ (b) $(13B57)_{16}$ (c) $(22A56)_{16}$ (d) $(123A6)_{16}$
14. The addition of octal numbers, $(2713)_8 + (2135)_8$ is equal to:
    (a) $(4050)_8$ (b) $(5021)_8$ (c) $(5061)_8$ (d) $(5050)_8$
15. Solve this binary problem: 10000000 + 00001111
    (a) 10000000 (b) 01111111 (c) 10000001 (d) 01110001
16. Solve this binary problem: 10110110 + 01111000
    (a) 10001110 (b) 100101110 (c) 100001100 (d) 111100110

17. When subtracting $(010110)_2$ from $(1011001)_2$ using 2's complement, we get:
(a) 0111001 (b) 1100101 (c) 0110110 (d) 1000011

18. For a number system of base r, the digits which are allowed in its representation are?
(a) Digits from 1 to r (b) Digits from 0 to $r-1$
(c) Digits from 1 to $r-1$ (d) Digits from 0 to 2^r

19. Which numbering format is the basis for digital electronic circuitry?
(a) Decimal (b) BCD (c) Octal (d) Binary

20. By using DeMorgan's theorem, $X = (A'(B' + C))'$ is simplified to:
(a) $X = A(B + C)$ (b) $X = A + B + C'$
(c) $X = ABC'$ (d) $X = A + BC'$

21. If a three-switching function is expressed as the product of maxterms by:

$$F(A,B,C) = \prod(0,3,5,6)$$

Then, it can be expressed as the sum of minterms by:
(a) $\sum(0,3,5,6)$ (b) $\sum(1,2,4,7)$ (c) $\sum(1,2,3,7)$ (d) $\sum(0,1,2,3,7)$

22. A four-switching function:

$$F(A,B,C,D) = A'B'CD + A'BC'D + A'BCD + AB'C'D + AB'CD$$

Expressed as the sum of minterms by:
(a) $\sum(1,3,5,7,9)$ (b) $\sum(3,5,7,9,11)$ (c) $\sum(3,5,9,11,13)$ (d) $\sum(3,7,9,11,13)$

23. The four-switching function:

$$F(A,B,C,D) = \sum(5,9,11,14)$$

can be expressed as follows:
(a) $A'BC'D + AB'C'D + AB'CD + ABCD'$ (b) $A'BC'D + ABC'D + ABC'D + ABCD'$
(c) $ABCD + AB'CD' + AB'CD + A'BCD$ (d) $A'BCD + AB'C'D + AB'CD' + ABCD'$

24. The exclusive-OR (XOR), denoted by the symbol $\oplus$, is a logical operation that performs the following Boolean operation:
(a) $x \oplus y = xy' + x'y$ (b) $x \oplus y = xy + x'y$
(c) $x \oplus y = xy + x'y'$ (d) $x \oplus y = (xy') * (x'y)$

25. The exclusive- NOR is a logical operation that performs the following Boolean operation:
(a) $xy' + x'y$ (b) $xy + x'y'$ (c) $y + xy'$ (d) $(xy') * (x'y)$

26. The exclusive-OR (XOR) is equal to 1:
(a) $x = 1, y = 1$ (b) $x = 1, y = 0$

27. The exclusive-OR (XOR) is equal to 0:
(a) $x = 1, y = 1$ (b) $x = 1, y = 0$

28. Based on the principles of Boolean algebra, the simplest form of $X = A(B + C) + C$ is:
(a) $X = AB + AC$ (b) $X = AB + C$ (c) $X = A + B + C$ (d) $X = AB + AC + C$

29. Based on the principles of Boolean algebra, the simplest form of $X = AC + C'(A + B)$ is:
(a) $X = C' + AC$ (b) $X = AC + BC'$ (c) $X = A + BC'$ (d) $X = AC + B$

30. By using DeMorgan's theorem, $X = (A'(B' + C))'$ is simplified to:
(a) $X = A(B + C)$ (b) $X = A + B + C'$
(c) $X = ABC'$ (d) $X = A + BC'$

With my best wishes

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Answer	d	a	b	d	c	c	a	a	b	d	c	c	a	d	d	b	d	b	d	d	b	b	a	a	b	b	a	b	c	d